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PlayList Name : CORE JAVA FULL COURSE

Channel: India Free Internship

Live Notes with Live coding

NotesLink:
https://github.com/indiafreeinternship/NOTES/tree/main/core%20java%20notes

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SESSION 1

Full Stack Java Developer

Module 1-: Programming Module (CoreJava, AdvJava, Framework-Spring)

Module 2-: UI Module (Html/CSS/JS/Angular/React)

Module 3-: DataBase Module (Oracle,MySql)

Module 4-: Testing Module (Testing basics: Selenium)

Module 5-: Tools Module (DevSecOps: Development Security and Operations tools)

JAVA

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PART 1: Core Java

PART 2: AdvJava

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CORE JAVA

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Language

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=> Every language will have their own Alphabets.

=> Every lauguage will have their own Grammer.

=> Every language will have their own Constructions rules.

part 1: CoreJava

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1. Java Programming Components(java, Alphabets)

2. Java Programming concepts

3. Object Oriented Programming features.

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1. Java Programming Components(java, Alphabets)

(a) Variables

(b)Methods(Functions)

(c) Blocks

(d)Constructor

(e)classes

(f)interfaces

(g) AbstractClasses

2. Java Programming concepts

(a)Object-Oritented progrmming concepts

(b) Exception Handling process

(c) Multi-Threading process

(d) Java Collection Framework(JCF)

(e) File Storage

(f) Networking in java(Socket programming)

3. Object Oriented Programming features.

(a) Class

(b) Object

(c) Abstraction

(d) Encapsulation

(e) PolyMorphism

(f) Inheritance

Define StandAlone Applications :-

=>The application which are installed in one computer and performs actions in the same computer are known as Stand Alone application or DeskTop Application or Window applications.

=> According to developer StandAlone application means,

No Html Input

No Server Envrironment

No Database Storage

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AdvJava

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=> Advjava provide the following technolgies to construct web application.

(I)JDBC

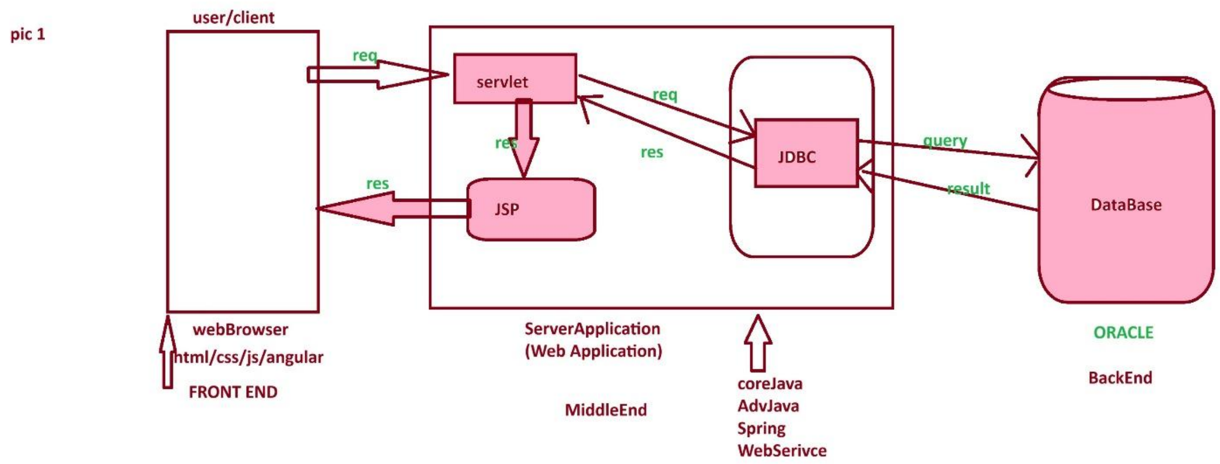
(ii)Servlet

(iii)JSP

Define Web Application

=>The Application which is executing in Web environment or internet environment is known as Web application or internet application.

Web Application Architecture pic 1:



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Servlet :- Servlet is a server program which accepts requets from user through WebBrowser.

JSP :- JSP stands for 'java Sever Page' and which is response from webApplication.

JDBC :- JDBC stands for 'java Database connectivity' and which is used to intract with Database.

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SESSION 2

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What is the differenct b/w

(i) Language

(ii) Technology

(iii) Framework

(i) Language = Language provide programming components and concepts which are used in contructing program.

eg: java,python,c,c++

(ii) Technology = The process of Converting knowledge into realtime world application development is known as Technology.

eg: Login, Registration, AdvJava.

(iii) Framework = The Stucture which is ready constructed and available for application development is known as Framework.

eg : Spring, webServices

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Module 1-: Programming Module (CoreJava, AdvJava, Framework-Spring)

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Level -1: core java - standAlone Application

Level -2 : AdvJava - Web Application.

Level -3 : Framework - Enterprise application.

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what is Enterprise application?

The application which are executing in distributed evironment and depending on the features like "Security", "Load balancing" and "clustring", are known as Enterprise Distributed Application or Enterprise Application.

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session 3

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Define Program?

- Program is a set of instruction.
- After writing program, we have to save the program with language extension.

eg: Test.c

Test.cpp

Test.java

Stages of Program:

- The program will have following two stages:

1. Compilation
2. Execution

Compilation:

- The process of checking the program constructed according to the rule of language or not, is known as Compilation process.
- After Compilation process is Successfull, the sourceCode is converted into Compiled codes.
- C and C++ programs will generate ObjectiveCode and java program generate Byte code.

Execution

- The process of running Compiled codes and checking the required output is generated or not, is known as Execution process.

- In C and C++ languages, while executing Objective Code Internally " loading and Linking" process is performed and Object code converted into excutable code and generate result.

- In Java Language, the Byte code is executed directly on JVM (Java Virtual Machine)

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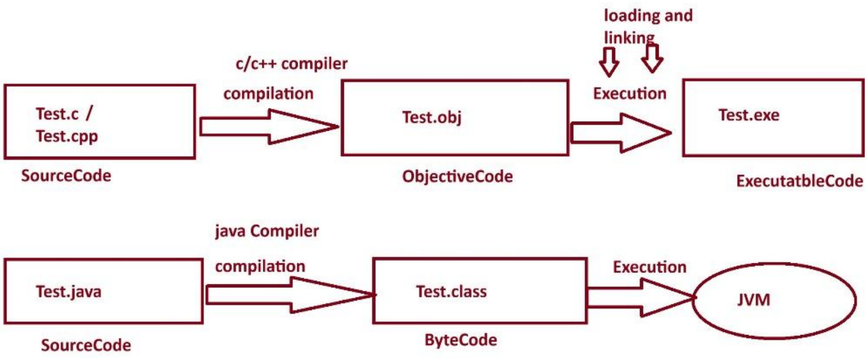
stages of program pic 2.

PIC 2



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session 4

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what is the diff b/w

(i) Objective Code

(ii) Byte Code

(i) Objective Code

==============

=> The copiled code genreated from c and c++ program is known as Objective Code.

=> While Objective code generation OperatingSystem is participated, becuase of this reason ObjectiveCode is PlatForm dependent code.

DisAdvantages

=> Objective Code generated from one PlatForm cannot be executed on other PlatForm.

Note:

=> c and c++ language which are generating Objective Code are PlatForm dependent language.

(ii) Byte Code

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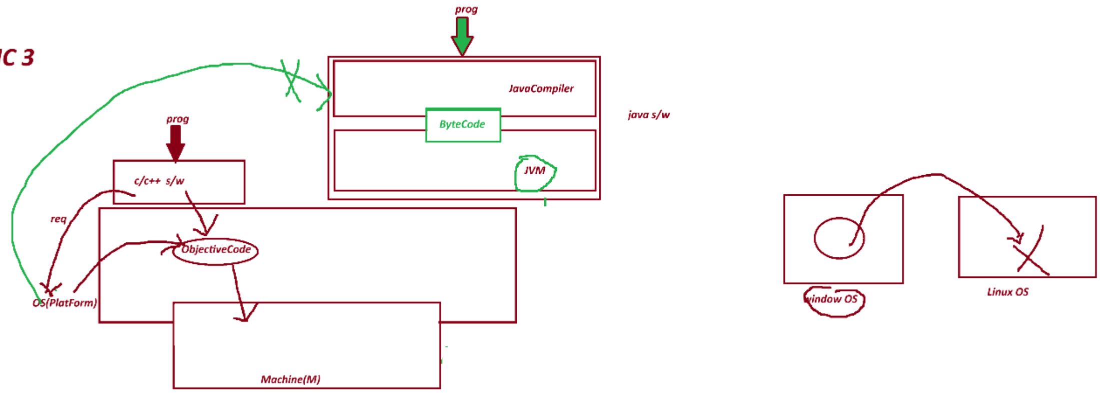
=> The byte code generated from one PlatForm canbe executed on all PlatForm based on JVM.

Note:

=> java language which is generating ByteCode is PlatForm independent language.

PIC 3

PIC 3



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Define High Level language?

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=> The language format whic are understandable by the users are known as High Level Language.

c,c++,java

Define Low Level Language?

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=> The language format which are not understandable by the user are known as Low Level Language.

Ex: Machine languages.

Define Translator?

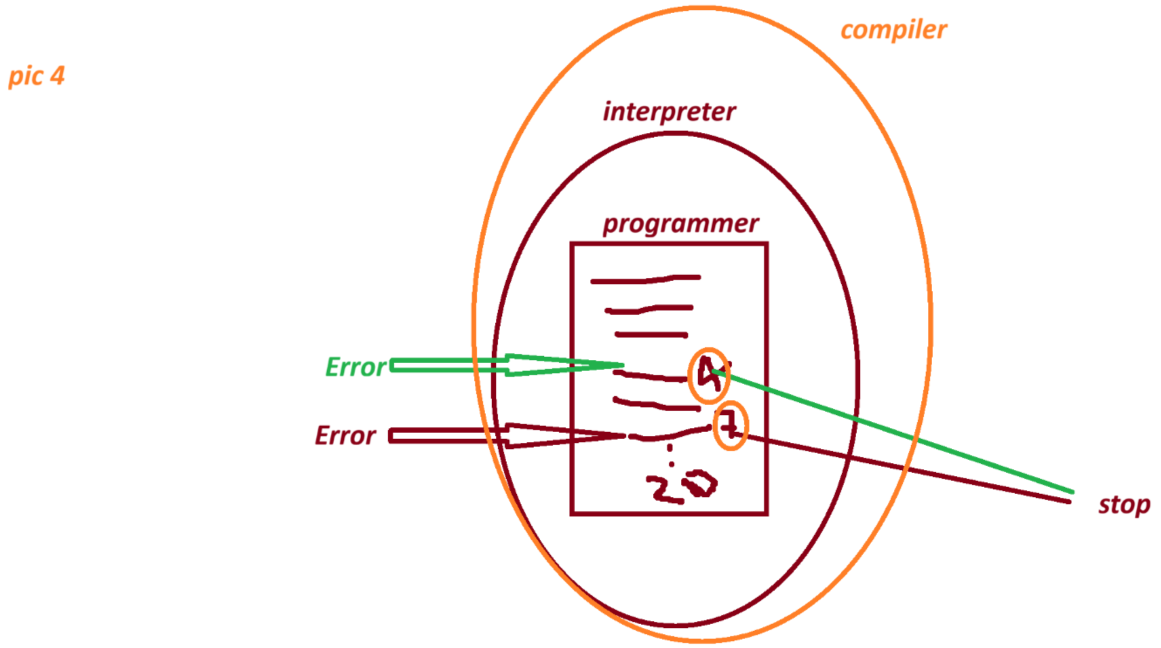
=> Translator are used to convert High Level language formats into Low Level Language formats and Low Level Language formats into High Level Language.

These Translators are categorized into two type.

(i) Interpreters - which translates program line-by-line.

(ii) Compiler - which translates total program at-a-time.

Pic 4



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1991 - "James Gosling" Sun Micro System - Code Write(programer)

WORA - Write Once and Run Anywhere.

Test.gt (Green Talk)

OAK

Java

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java versions:

1995 - java Alpha&Beta

1996- JDK 1.0

1997- JDK 1.1

1998 - JDK 1.2

2000 - JDK 1.3

2002 - JDK 1.4

2004 - Java5(Tiger)

2006 - java6

2011 - java7

2014- java8

2017 - java9

2018-java10 , java11

2019- java12,java13

..... 2026 -java 26

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session 5

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What is the diff between ?

(i) JDK

(ii) JRE

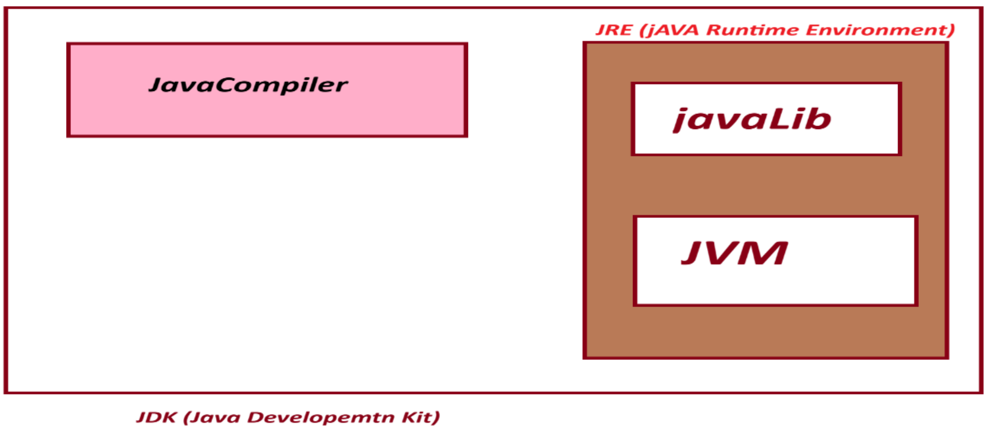
(iii) JVM

(i) JDK : JDK stands for "Java Development Kit" and which is the collection of the following:

(a) JavaCompiler

(b) JavaLib

(c) JVM



(a) JavaCompiler :- JavaCompiler will compile the source code and generate ByteCode when the compilation process is successfull.

(b) JavaLib :- JavaLib will provide pre-defined and ready constructed programming components which are used in construting application.

JavaLib= 'java'

CoreJava Packages:

(i) java.lang - Language package

(ii) java.util - Utility package

(iii) java.io - IO Stream and Files packages

(iv) java.net - Networking package

AdvJava packages:

(i) java.sql - DataBase connection package

(ii) javax.servlet - Servlet progrmming package

(iii) javax.servlet.jsp - JSP programming package

(c) JVM :- JVM Stands for "Java Virtual Machine" and which is used to execute JavaByteCode.

=> JVM internal partition of JDK.

=> The s/w components which intrnally having the behaviour like machine is known as Virtual Machine.

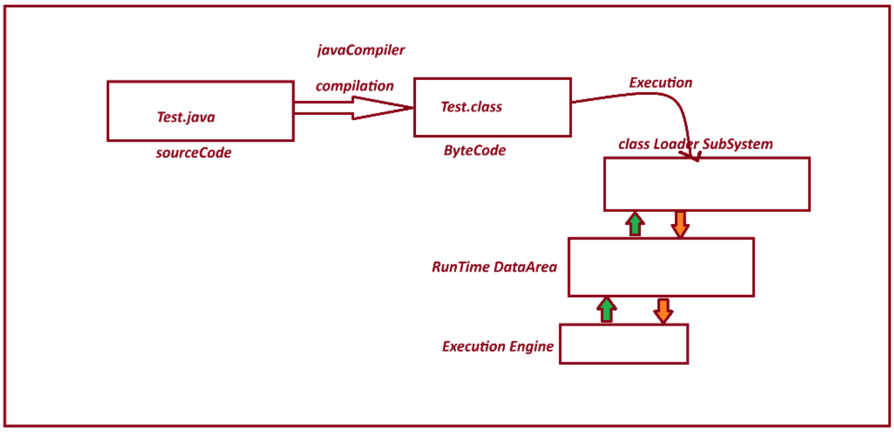
=> The Following are the parts of JVM:

1. Class Loader SubSystem

2. Runtime DataArea

3. Execution Engine

JVM PIC



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(ii) JRE :- JRE Stands for 'Java Runtime Environment' and which is collection JavaLib and JVM.

=> JRE is an internal partition of JDK.

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session 6

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Installing Java s/w and setting path:

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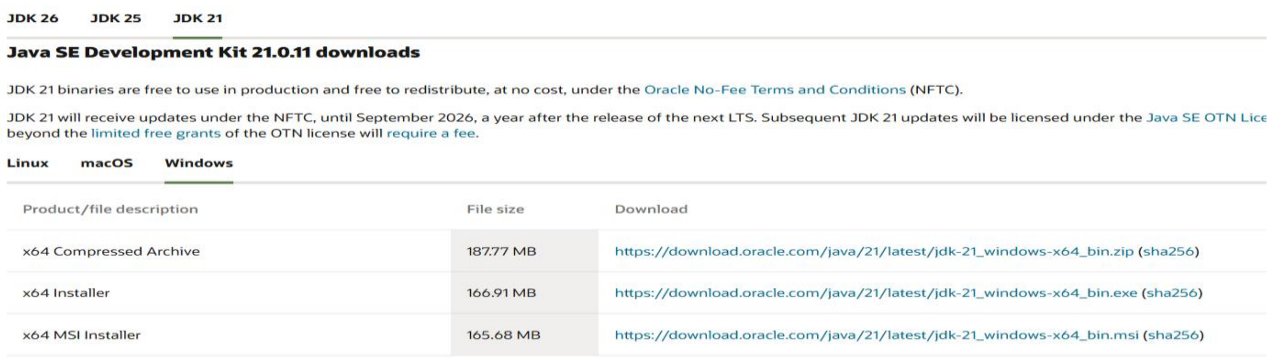
step 1 :- Download java s/w(JDK) from Oracle Website.

Link: https://www.oracle.com/java/technologies/downloads/

step-2 : Install JDK-21



JDK 26 JDK 25 JDK 21

Java SE Development Kit 21.0.11 downloads

JDK 21 binaries are free to use in production and free to redistribute, at no cost, under the Oracle No-Fee Terms and Conditions (NFTC).

JDK 21 will receive updates under the NFTC, until September 2026, a year after the release of the next LTS. Subsequent JDK 21 updates will be licensed under the Java SE OTN Lice beyond the limited free grants of the OTN license will require a fee.

Linux macOS Windows

Product/file description	File size	Download
x64 Compressed Archive	187.77 MB	https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.zip (sha256)
x64 Installer	166.91 MB	https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.exe (sha256)
x64 MSI Installer	165.68 MB	https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.msi (sha256)

Note: After Installation process we can find one folder with name "java" in program file.

C:\Program Files\Java

step 2 : set java path in "Envrironment variables"

click->new->

variables : path

variable : C:\Program Files\Java\jdk-21.0.11\bin;

Note:

=> Open CommandPrompt and check the follwoing commands are working or not:

javac - compilation commands

java -execution command

C:\Users\satru>java -version

java version "21.0.11" 2026-04-21 LTS

Java(TM) SE Runtime Environment (build 21.0.11+9-LTS-211)

Java HotSpot(TM) 64-Bit Server VM (build 21.0.11+9-LTS-211, mixed mode, sharing)

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session 7

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Environment Variables :- Envrironment Variables are OperatingSystem variable, which hold the information about s/w components installed in Computer System.

These Environment variable are categorized into two types.

(i) User variable

(ii) System Variables

(i) User variable :- The variable whihc are related to individual users of Computer System are known as User Variables.

=>The information in User variable can be used by only individual user.

(ii) System Variables :-The variable which are related to all the users of computer system are known as system variables.

=> The Information in System variable can be used by all users of computer system.

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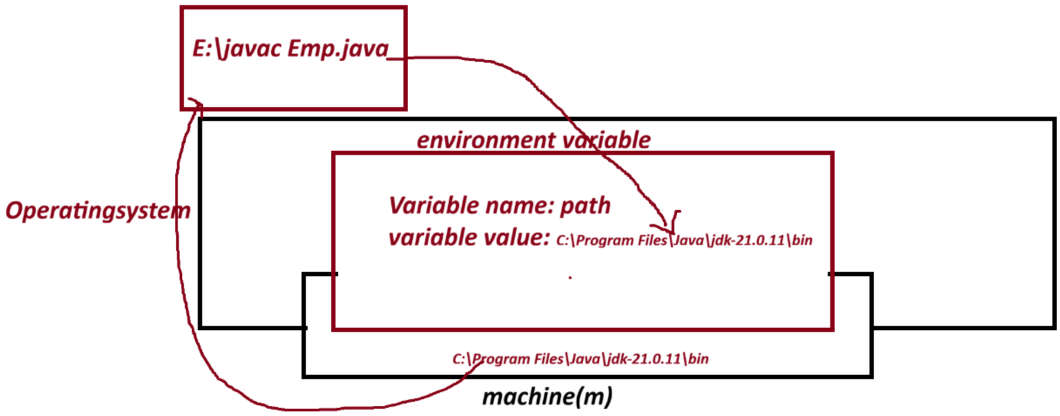
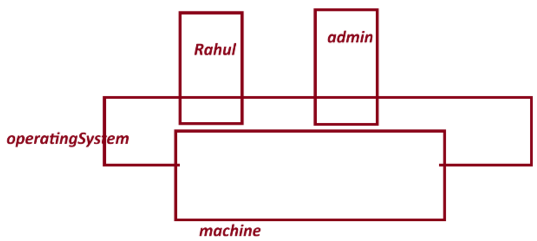
what is the advantage of having javaPath in environment variable?

=> When we have javaPath in Environment variable, then we can compile and execute java program from any location of computer System.

diagram:





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session 8

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define 'class'?

=> "class" is a "Structured Layout" generate "Object".

=> class in java is a collection of Variables, Methods, Blocks and Construtors.

=> The class in java can be added with main() to start the program execution.

=> main() is having the following pre-defined standard systax

ex: public static void main(String[] arg){}

=>we use "class" keyword in java to construct classes.

Structur of class in java:

```
class Class_name{
        //variable
        //methods
        //Blocks
        //Construtors
        //main()
}
```

===================================

Ex program-1 : - Wap to display the msg as "Welcome to India Free Internship"?

```
class Display{
public static void main(String[] args){
 System.out.print("India Free Internship");
  }
}
```

step -1 Open Notedpad and type the program.

step -2 Save the program in folder with language-extendion (.java)

Systax :

Class_name.java

ex:

Display.java

step 3- File->Save-> Browser and select folder -> core java prgrams

-> save as type must be "All Files" -> click "save"

=> Open CommandPrompt and perform compilatin and Execution process

To open CommandPrompt , goto,folder-> type "cmd" in Addressbar and press "Enter"

step 4- compile the program as follows:

Syntax:

javac Class_name.java

ex:

javac Display.java

Step -5 Execute the program as follows:

Syntax:

java Class_name

Ex:

java Display

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Session 9

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Ex program -2

wap to display the sum of two numbers?

```
class Addition{
                public static void main(String[] args){
    int a=10;
```

```
    int b=20;
    int c=a+b;
    System.out.println("a value = "+a);
    System.out.println("b value = "+b);
    System.out.println("result (a+b) ="+c);
  }
}
```

ouput

=====

D:\core java programs>javac Addition.java   (compilation)

D:\core java programs>java Addition (execution)

a value = 10

b value = 20

result (a+b) =30

===============

what is the diff b/w print() and println()

(i) print()

(ii)println()

(i)print():- print() method will display message and result, and makes the cursor wait in the same line.

(ii) println():- println() method also display message and result, and moves the cursor to next line or new line.

Note:- "+" symbol in print() method will concatenate(combine) message with result.

==

Ex - program-3

wap to display Employee details?

(id,name,desg,sal)

Employee.java

```
class Employee
{
   public static void main(String[] args)
   {
     String id="SE121";
     String name="Alex";
     String desg="SE";
     int sal=30000;
     System.out.println("EmpId ="+id);
     System.out.println("EmpName ="+name);
     System.out.println("EmpDesg ="+desg);
     System.out.println("EmpSalary ="+sal);
   }
}
```

ouput

```
javac Employee.java
java Employee

EmpId =SE121
EmpName =Alex
EmpDesg =SE
EmpSalary =30000
```

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Assignment -1

wap to display BookDetails?

(code,name,author,price,qty)

Assignment -2

wap to display StudentDetails?

(name,branch,rollNo,6 sub marks, totMarks,per)

ouput

------

name=

branch=

rollNo=

s1=

s2=

s3=

s4=

s5=

s6=

totMarks=s1+s2+s3+s4+s5+s6;

per=

====================

session 10

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Naming Conventions in java.

=>The coding rules which are followed by the progrmmer in writing porgrams are known as Naming conventions in Java.

(Realtime Coding Standard).

package:

def => Packages are collection of Classes,interfaces and enum.

rule=> packages must be writing in Lowercase.

ex. com.app

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Classs and interfaces

def => Classes and Interface are collection of "variable and Methods"

rule : In Classes and interfaces, the starting letter of every word must be UpperCase.

ex: EmployeeSalary, DataInputStream

===========================

Variables and Methods:

def : Variables are data holders and Methods are actions.

rule : In Variables and methods, the first word must be in LowerCase and from second onwords the starting letter must be UpperCase.

Ex: variable : rollNo, panCard,aadharCard

methods: readLine(), getSalary(),calculateMakrs(),.. etc

==

Keywords:

def : The pre-defined built in words are known as keyword.

Rule : Keywords in java must be LowerCase.

Ex: void, static, public...